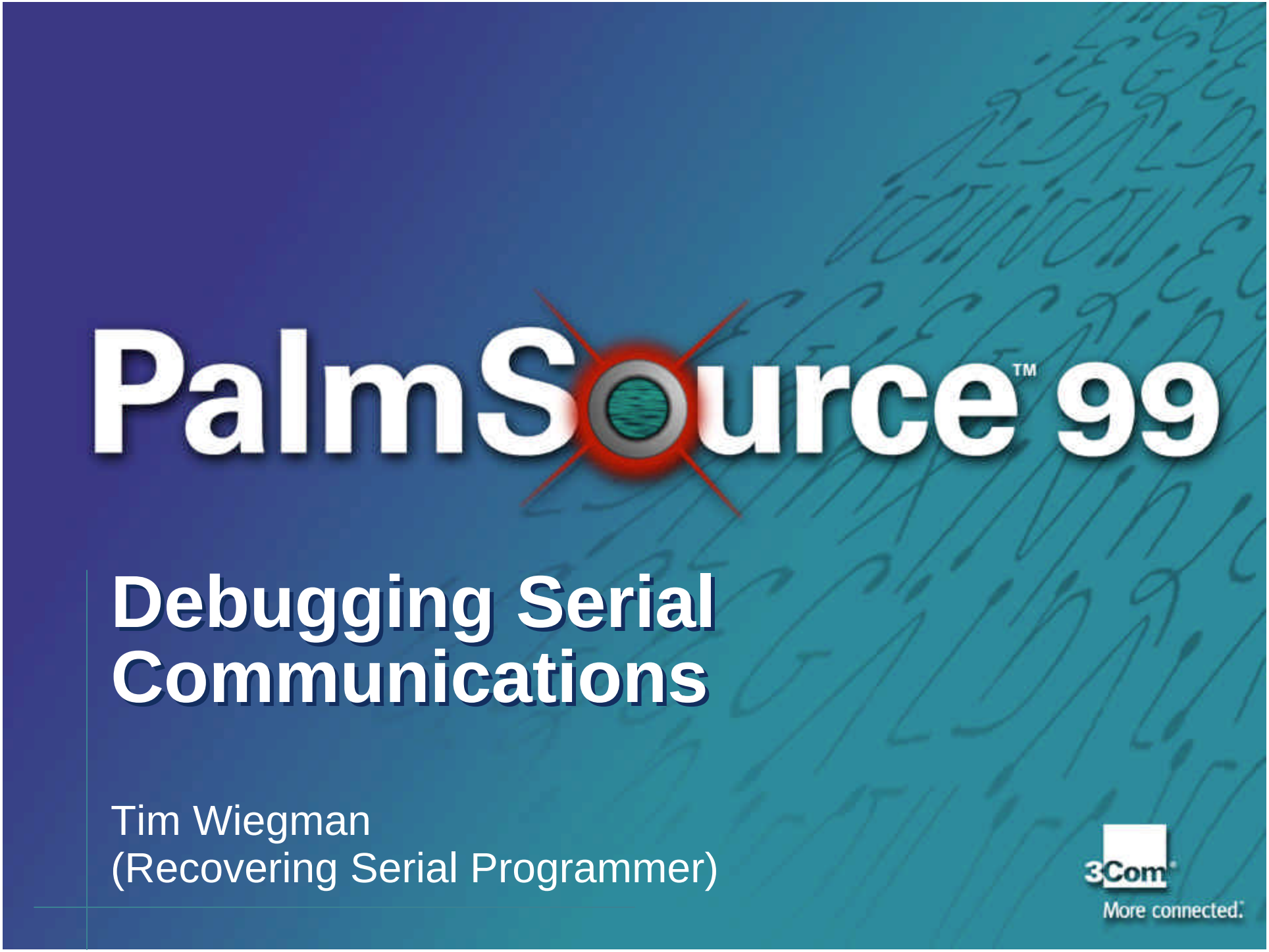

PalmSource™ 99
Debugging Serial Communications
Tim Wiegman
(Recovering Serial Programmer)
3Com
More connected.

Overview

PalmSource 99

- Why not use a source level debugger?
- General tips
- Alternate debugging methods
- Palm Debugger
- Palm OS® Emulator
- Debugging examples

Why Not Use a Source Level Debugger?

PalmSource 99

- Real-time nature of most protocols prevents stopping execution for long periods of time
- Source level debugger might work for direct serial connection with simple protocols

General Tips

- Test your application using simple serial communication device first
  - For example, if writing IR app, test using wired serial first then introduce IR
- Make sure you are cabled correctly
  - Remember that HotSync® cradle cable is already wired as a null-modem cable
  - Use PC terminal emulator or Palm OS® Emulator to verify data is arriving

Alternate Debugging Methods

PalmSource 99

- Printf
- Audio cues
- Writing to the screen
- Asserts
- Hardware
- Logging

Printfs

- DbgMessage sends text messages to the debug console
  - Conflicts with serial port your app is using
  - Slow when used with Palm OS Emulator
- Use HostFPrintF when debugging using Palm OS Emulator

Audio Cues (aka ‘Beeps’)

PalmSource 99

- SndPlaySystemSound(sndError)
- Usually not a great way to debug
- Might be useful for SrmSetWakeupHandler callback

Writing to the Screen

PalmSource 99

- Have an area on the display reserved for status/error info
- Can get busy if too many of these status areas are used
- Lose context if a lot of information is being overwritten to the same area

”Asserts"

PalmSource 99

- ErrNonFatalDisplay
- Good way to catch unexpected situations

Hardware

- Second serial port for debugging
  - PalmPilot upgrade card has 2nd serial port, so serial port can be used for debugging if application uses IR
  - Think of Palm OS Emulator as a second serial port
- Break-out box
  - Poor man’s serial analyzer
- Serial analyzer
  - Nice if you have one

Logging

- Use conditional macros so logging info can be selectively chosen based on problem
- Log to circular RAM buffer
  - Otherwise writing to database interferes with serial communication timing
- Write log to memo pad entry(s) at close of serial device
- Alternatively, write log to HotSync log
- Palm OS Emulator has built-in serial logging

PalmDebugger w/Real Device

PalmSource 99

- Connect to debugger before testing
  - Use shortcut . 1
  - Prevents hangs in ErrorDisplay dialog
- Connect to debugger after debugger is called (or crash occurs)
  - May have to change connection speeds to match serial device speed to successfully attach to debugger

Palm OS Emulator—Serial Port/IR Emulation

- Palm OS Emulator emulates serial port
- IrLib I/O redirected thru serial port when running on Palm OS® Emulator
  - Has trouble if receiving connection (timing issues)
  - Works okay if initiating connection
- Have to configure receiving device to use IrDA protocol through serial port

PalmDebugger w/Palm OS Emulator

- Doesn't use serial to communicate with Palm OS Emulator, so doesn't get in way of serial driver
- Can test (and debug fatal situations) without the need for a real device

Palm OS Emulator—HostControl"

- Another way to do logging
- May not be appropriate for all situations
  - Some things require real device(Ir, Palm VII)

Header Files for Info Mentioned

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- (All in Incs/Core/System)
- DLServer.h
- ErrorMgr.h
- HostControl.h

Debugging Examples…

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- Using real device
- Using Palm OS Emulator


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